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INDUSTRIAL PRACTICAL TRAINING REPORT

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DECLARATION

I sincerely declare that this project work entitled "INDUSTRIAL PRACTICAL TRAINING REPORT", submitted to the National Institute of Transport supervisor towards the partial fulfillment of the fieldwork , is a record of the original work done by me under the guidance of Mr. Frezier Yahaya (the head of department) and Mr. David Chonya (the supervisor), of the Company Datahouse Tanzania Limited. The details of training and experience contained in this report describe my involvement as a trainee in the field of computer science.

Submitted by:

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TABLE OF CONTENTS

Ch. No.	Topic	Page No.
	Student Declaration	1
	Table of Contents	2
	Abbreviations	3
1	**Introduction** 1.1 Background 1.2 Organization Structure 1.3 Vision ,Mission ,Objective 1.3.1 Vision 1.3.2 Mission 1.3.3 Objectives	4-6
2	**Activities Performed** 2.1 Overview 2.2 Activities performed during IPT 2.3 General Observations 2.4 Challenges and How challenges were solved	7-9
3	**Conclusion and Recommendations** 3.1 Conclusion 3.2 Recommendations	10-11
	References	12
	Appendix	13

ABBREVIATIONS

●NIT	National Institute of Technology
●AI	Artificial Intelligence
●DHTL	Datahouse Tanzania Limited
●IT	Information Technology
●ICT	Information Communication Technology
●BI	Business Intelligence
●ERP	Enterprise Resource Planning
●CRM	Customer Relationship Management
●GIT	Global information tracker

CHAPTER 1

Introduction

1.1 Background:

Datahouse Tanzania Limited was found in 2006 and initially started offering basic IT solutions by 2007.Through the year 2008 to 2010 Datahouse shifted its focus towards business process automations, developing expertise in document life cycle solutions.

From 2011 to 2013, Datahouse launched tailored document management solutions in Tanzania, witnessing the amount of increased efficiency for clients and expanded its solutions in 2014, becoming a leader in business process automation within Tanzania.

In 2017, Datahouse Expanded to Rwanda, introducing document life cycle solutions tailored to the market there and then later on entered Uganda, solidifying its reputation for streamlined document management across borders by 2018.

By 2020-2021, Datahouse was recognized in most parts of East Africa as in being recognized as the preferred provider of document life cycle solutions in East Africa.

As for the year 2023/2024 Datahouse continues its growth, focusing on AI and machine learning for enhanced document automation.

1.2 Organization Structure:

Datahouse Tanzania Limited is comprised of a lot of employees, the average company size is 11 to 50 employees. The founder and managing director of Datahouse Tanzania Limited is Godfrey Woiso. The other employees include the software engineers that help other companies in the digital transformation processes.

The company also comprises of different workers working at the sites since some of the company's work is practical. Datahouse Tanzania Limited has a well-structured organizational framework designed to foster efficiency and collaboration. At the top is the executive management team, responsible for strategic decision-making and company direction. Below them are department heads overseeing various functions such as IT, data analytics, operations, and human resources. Each department is composed of specialized teams working together to achieve project goals.

The structure encourages cross-departmental collaboration, ensuring a smooth workflow from top management to site workers, enabling DHTL to deliver practical and effective solutions. A lot of the Senior Subject Matter Experts and System Designers at DHTL have extensive ICT experience and have been within the ICT industry for over ten years, bringing stability and continuity as they have grown to prominence in the industry.

There are developed implementations, as well as development and support processes, designed to maximize efficiency and productivity. DHTL teams are thoroughly trained in the use of these techniques.

Exceptional functional and technical expertise coupled with extensive industry knowledge makes DATAHOUSE, the ideal choice for managing many ICT projects.

1.3 Vision, Mission, Objective:

- **Vision** is the long-term aspiration of where the organization wants to be in the future.

- **Mission** describes the organization's current purpose, what it does, and how it delivers value.

- **Objectives** are specific, actionable steps taken to achieve the mission and ultimately fulfill the vision.

1.3.1 Vision

Datahouse aspires to be the leading provider of innovative data-driven solutions in Tanzania, empowering businesses and institutions with actionable insights for sustainable growth and development.

1.3.2 Mission

DHTL's mission is to deliver reliable, efficient, and cutting-edge data solutions that enhance decision-making for the clients. Through advanced technology, expert knowledge, and a customer-focused approach, DHTL aims to drive digital transformation and foster economic progress across Tanzania.

1.3.3 Objectives

1. **Deliver high-quality data services** that meet the evolving needs of businesses and organizations.
2. **Foster innovation** by continuously integrating the latest technology and methodologies into their solutions.
3. **Enhance customer satisfaction** by maintaining excellent service standards and ensuring timely delivery of projects.
4. **Promote sustainable growth** by leveraging data to empower businesses in making environmentally conscious and socially responsible decisions.

CHAPTER 2

Activities Performed

2.1 Overview:

Datahouse Tanzania Limited is a Tanzanian-based company that provides specialized IT services and data solutions for businesses across various industries. The company's focus is on delivering technology-driven solutions to help organizations optimize their operations, make data-driven decisions, and improve their overall efficiency.

The Key Services include;

1. **Data Solutions:** Datahouse specializes in offering comprehensive data management services, including data storage, analytics, and visualization to enable businesses to utilize their data for strategic decision-making.
2. **Software Development:** The company offers custom software development services, tailoring solutions to meet the specific needs of clients. This includes enterprise software, mobile applications, and web development.
3. **IT Consultancy:** Datahouse provides consultancy services to organizations looking to integrate new technologies, improve their IT infrastructure, or enhance data security.
4. **Cloud Computing:** As cloud computing becomes more critical in business operations, Datahouse offers cloud solutions that include cloud migration, cloud storage, and cloud-based applications.
5. **Business Intelligence (BI):** The company assists businesses in harnessing the power of BI tools to gain insights from their data, allowing them to identify trends, opportunities, and potential risks.
6. **Digital Transformation:** Datahouse helps businesses transition into the digital age by offering solutions such as process automation, customer relationship management (CRM) systems, and enterprise resource planning (ERP) systems.

2.2 Activities performed during IPT

As a first year student there is a variety of general activities designed to help in gaining practical experience and a deeper understanding of the IT and data solutions industry. Here's a list of activities performed:

1. **Introduction to IT Infrastructure**

The primary goal being gaining a foundational understanding of the company's IT infrastructure, which included connecting devices, setting up workstations for employees, getting familiar with the physical computer equipment, and be able to assist with the setup, maintenance, and troubleshooting of hardware and software systems.

2. **Collaboration and Teamwork**

This main part involved the use of GIT where the sole purpose was to get involved with software development as a team since it is a free and open-source version control system used to handle small to very large projects efficiently. Git is used to tracking changes in the source code, enabling multiple developers to work together on non-linear development.

3. **Basic Software Development Tasks**

The main objective was to develop foundational coding skills and contribute to small-scale software projects. Working on frontend and backend development under supervision also engage in code reviews and testing software functionality and lastly use programming languages like Python, JavaScript,PHP, or HTML/CSS for simple projects.

4. **Assisting with easy Tasks**

Activities like assisting with basic administrative tasks such as scanning documents, printing materials, and arranging files. These tasks will helped to become familiar with office workflows and organization while supporting the team in maintaining efficient document management systems. Through these responsibilities, there is developed attention to detail and gain practical experience in handling everyday office operations.

2.3 General Observation

As a first-year trainee at Datahouse Tanzania Limited, there is exposure to a wide range of tasks that bridge both administrative and technical roles. From assisting with basic administrative tasks like scanning, printing, and file arrangement, to gaining hands-on experience with IT infrastructure, the opportunities are designed to help in building foundational skills. The activities mainly ranged from learning about hardware setup and network configurations, to supporting software development and handling computing projects.

Also being introduced to essential aspects of data management, business intelligence, and digital transformation, providing knowledge with a comprehensive understanding of how IT solutions are implemented in real-world scenarios. This diverse exposure, including security measures and client communication, contributing to a well-rounded learning experience.

Overall, the role as a trainee helped in developing both technical and soft skills, setting a strong foundation for a future career in the IT and data solutions industry.

2.3 Challenges and Solutions

- The wide range of IT tools, software, and technical matters may feel overwhelming, especially when being new to the field. The issue was solved by focusing on mastering the basics first. Breaking them down from complex concepts into manageable parts and seeking guidance from mentors and senior staff. Taking advantage of the available training resources and online tutorials provided by the company to gradually build my knowledge.

- Tasks like troubleshooting technical issues and configuring hardware came to be challenging due to a lack of hands-on experience. The matter was minimized by observing the senior staff as they work through these tasks. Being proactive in asking questions and practicing whenever possible and having a way to know how to communicate using the technical information clearly for a much better understanding.

CHAPTER 3

Conclusion and Recommendations

1. Conclusion:

As a first-year trainee at Datahouse Tanzania Limited, I was able to encounter a diverse range of tasks that offered both learning opportunities and challenges. From administrative duties to technical responsibilities in IT infrastructure, software development, and data management, the role provided a strong foundation for developing essential skills. However, managing the technical learning curve, balancing tasks, and adapting to a fast-paced environment did require persistence and proactive learning. By overcoming these challenges with the right approach, I was successful to gain some valuable experience that shaped my career startup in the IT and data solutions industry.

2. Recommendations:

Developing a structured learning plan: To address the technical learning problem, creating a personalized learning plan that focuses on specific areas of weakness, such as networking, cloud computing and software development. Breaking the plan into smaller, manageable parts, and tracking my progress over time.

Seeking mentorship and Guidance: Taking advantage of the experience and knowledge of senior colleagues. Regularly ask for feedback and seeking mentorship to improve my skills. This will also help me to navigate complex technical tasks with more confidence.

Prioritizing time Management: Effective time management will be key to handling multiple responsibilities. Using task prioritization techniques to balance routine tasks with technical learning and project participations.

Embracing continuous Learning: Staying committed to ongoing learning by engaging in the IT industry, online courses, and reading relevant materials. This will help me stay current with evolving technologies and enhancing my problem-solving abilities.

Enhancing Communication Skills: Working on simplifying complex technical concepts. Regularly practicing my technical communication skills will improve my ability to collaborate effectively and build professional relationships.

Being adaptable and open to Feedback: Being flexible in adapting to the fast-paced environment. Accepting constructive criticism as a tool for growth and to remain open to adjusting my approach when new challenges arise.

References:

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●Association for Computing Machinery (ACM). (2022). "Guidelines for Professional Development in Computing."This document offers strategies and recommendations for continuous learning in the computing field. Available at: ACM.

●Coursera. (n.d.). "IT Infrastructure and Emerging Technologies.",An online course that introduces the key concepts of IT infrastructure and how emerging technologies impact business operations. Available at: Coursera.

●LinkedIn Learning. "Learning Network Infrastructure.",This resource offers video courses on network infrastructure, including practical insights into setup and management. Available at: LinkedIn Learning.

Appendix:

Common Tools and Software Used

- **Hardware:**
    - Dell Servers
    - Cisco Routers and Switches
- **Software:**
    - Windows Server
    - VMware for virtualization
    - Microsoft Office Suite for administrative tasks